

SASC on CRC data

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Load functions:

```
library(scry)
library(ggplot2); theme_set(theme_bw())
library(umap)
library(mvtnorm)
library(MCMCpack)
library(cluster)
library(salso)
library(ggpubr)
library(dplyr)
library(fdrtool)
library(Seurat)

source("SASC_func.R")
source("realdata_func.R")

data_dir <- "../data/realdata"
output_dir <- "../output/realdata"
output_data_dir <- "../output/realdata/data"
output_figure_dir <- "../output/realdata/figures"

# colors
pal <- c("#ffb6db", "#33A02C", "#b66dff", "#FEC44F", "#41B6C4", "#8E0152", "#0868AC", "#807DBA", "#E7298A",
         "#00441B", "#525252", "#4D9221", "#8B5742", "#D8DAEB", "#7cdd2d", "#980043", "#8C96C6", "#EC70C2",
         "#FDAE61", "#1D91C0", "#A6DBA0", "#4292C6", "#BF812D", "#01665E", "#41AB5D", "#FE9929", "#252525")
names(pal) <- 1:30
```

Read data:

```
if (file.exists(paste0(data_dir, "/CRC_final_annotation.rds"))){
  input <- readRDS(paste0(data_dir, "/CRC_final_annotation.rds"))
} else {
  cat("***Please first run 'preprocessing_CRC.Rmd' for data preprocessing***")
}

pdf(file = paste0(output_figure_dir, "/PCAelbow.pdf"), width = 5, height = 4)
ElbowPlot(input, ndims = 50)
dev.off()
```

```
## pdf
```

```
## 2
npc <- 25
input <- RunPCA(object = input, npcs = npc)

# count matrix
count <- as.matrix(input@assays$RNA@counts)
dim(count)

## [1] 12525 3139

# feature matrix after PCA (dim reduction)
gene <- input@reductions$pca@cell.embeddings
dim(gene)

## [1] 3139 25

# umap embeddings
gene_umap <- input@reductions$umap@cell.embeddings
dim(gene_umap)

## [1] 3139 2

# onset
onset <- input@meta.data$Onset
# cell names
cell_names <- colnames(count)
# CD8 CD4 type
T_type <- input@meta.data$T_cell_type
table(T_type)

## T_type
## CD4 CD8
## 1626 1513

annotation <- input@meta.data$annotation_final
table(annotation)

## annotation
## cd4T_hp cd4T_other cd4T_rg CD8T_em CD8T_ex CD8T_other
## 858 198 570 657 362 494

types <- names(table(annotation))
names(annotation) <- cell_names

# rename the groups to be L and E
onset_LE <- onset
onset_LE[which(onset == "LOCRC")] <- "L"
onset_LE[which(onset == "YOCRC")] <- "E"
```

Exploratory data analysis:

```
annotation_rename <- annotation
annotation_rename[which(annotation == "cd4T_hp")] <- "CD4T_hp"
annotation_rename[which(annotation == "cd4T_other")] <- "CD4T_other"
annotation_rename[which(annotation == "cd4T_rg")] <- "CD4T_rg"
Annotation <- annotation_rename
Onset <- onset
```

```

table_df <- table(Annotation, Onset)
pdf(paste0(output_figure_dir, "/mosaic.pdf"), width=6, height = 3.5)
mosaicplot(table_df,
            main = "Mosaic plot",
            color = TRUE
)
dev.off()

## pdf
## 2

fisher.test(table_df, simulate.p.value=TRUE)

##
## Fisher's Exact Test for Count Data with simulated p-value (based on
## 2000 replicates)
##
## data: table_df
## p-value = 0.0004998
## alternative hypothesis: two.sided

```

Visualize the data set:

```

pointsize <- 0.5

annotation_plot <- annotation
annotation_plot[which(annotation == "cd4T_hp")] <- "CD4+ T helper"
annotation_plot[which(annotation == "cd4T_other")] <- "CD4+ T other"
annotation_plot[which(annotation == "cd4T_rg")] <- "CD4+ T regulatory"
annotation_plot[which(annotation == "CD8T_em")] <- "CD8+ T effective memory"
annotation_plot[which(annotation == "CD8T_ex")] <- "CD8+ T exhausted"
annotation_plot[which(annotation == "CD8T_other")] <- "CD8+ T other"

pal0 <- pal[1:6]
names(pal0) <- annot_names <- c("CD4+ T helper",
                                "CD4+ T other",
                                "CD4+ T regulatory",
                                "CD8+ T effective memory",
                                "CD8+ T exhausted",
                                "CD8+ T other")

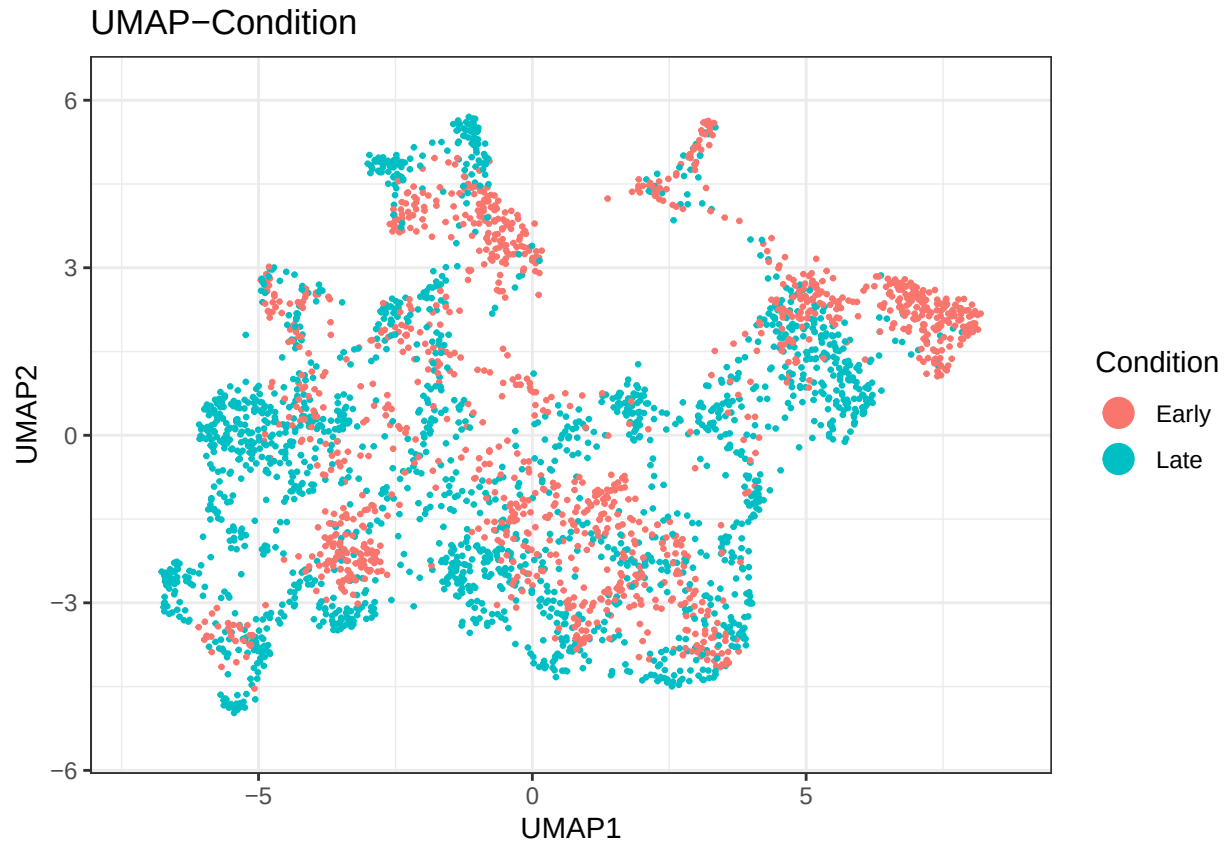
T_type_plot <- T_type
T_type_plot[which(T_type == "CD4")] <- "CD4+"
T_type_plot[which(T_type == "CD8")] <- "CD8+"
onset_LE_plot <- onset_LE
onset_LE_plot[which(onset_LE == "L")] <- "Late"
onset_LE_plot[which(onset_LE == "E")] <- "Early"
df <- data.frame(PC1 = gene[,1], PC2 = gene[,2],
                 UMAP1 = gene_umap[,1], UMAP2 = gene_umap[,2],
                 Condition = onset_LE_plot, T_type = T_type_plot, Annotation = annotation_plot)

xrange <- c(min(df$UMAP1) - 0.5 , max(df$UMAP1) + 0.5)
yrange <- c(min(df$UMAP2) - 0.5 , max(df$UMAP2) + 0.5)

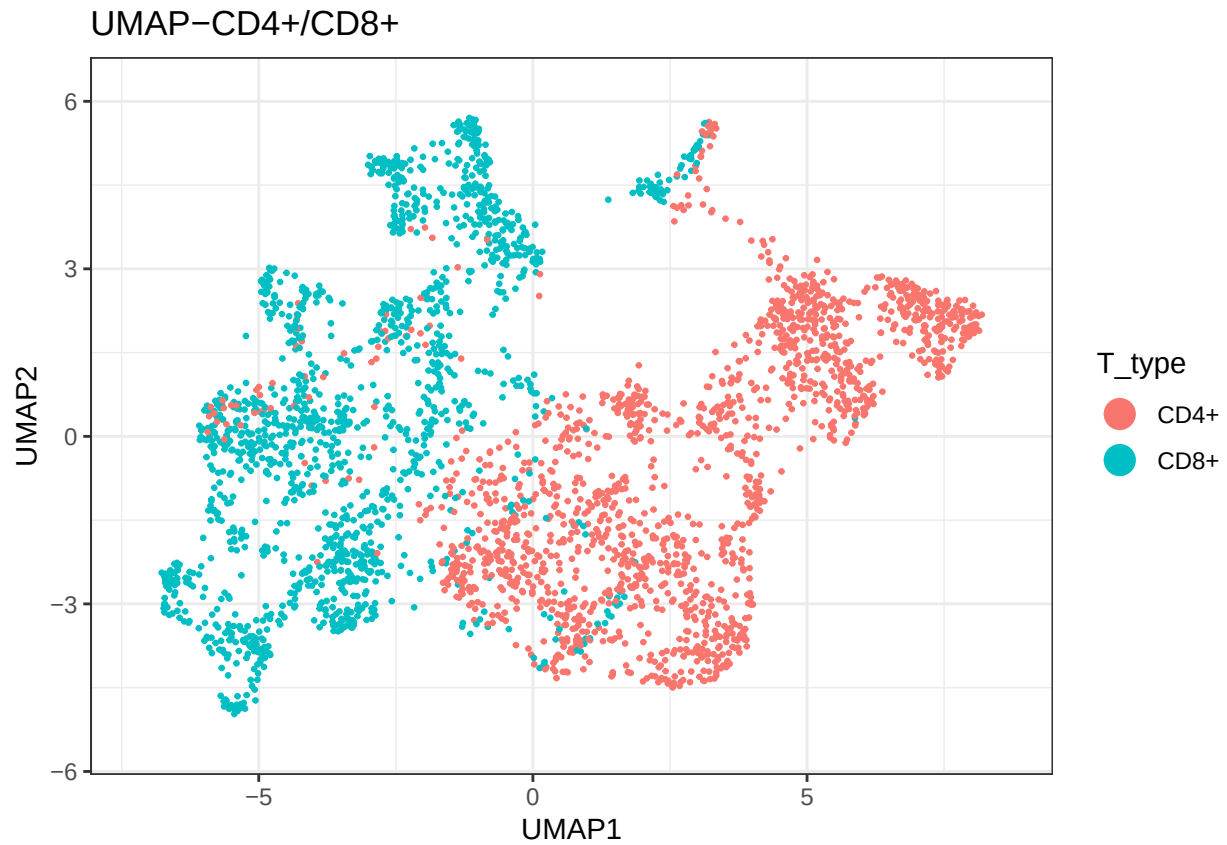
# plot

```

```
p1 <- ggplot(df, aes(x = UMAP1, y = UMAP2, color = Condition)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("UMAP-Condition")
p1
```

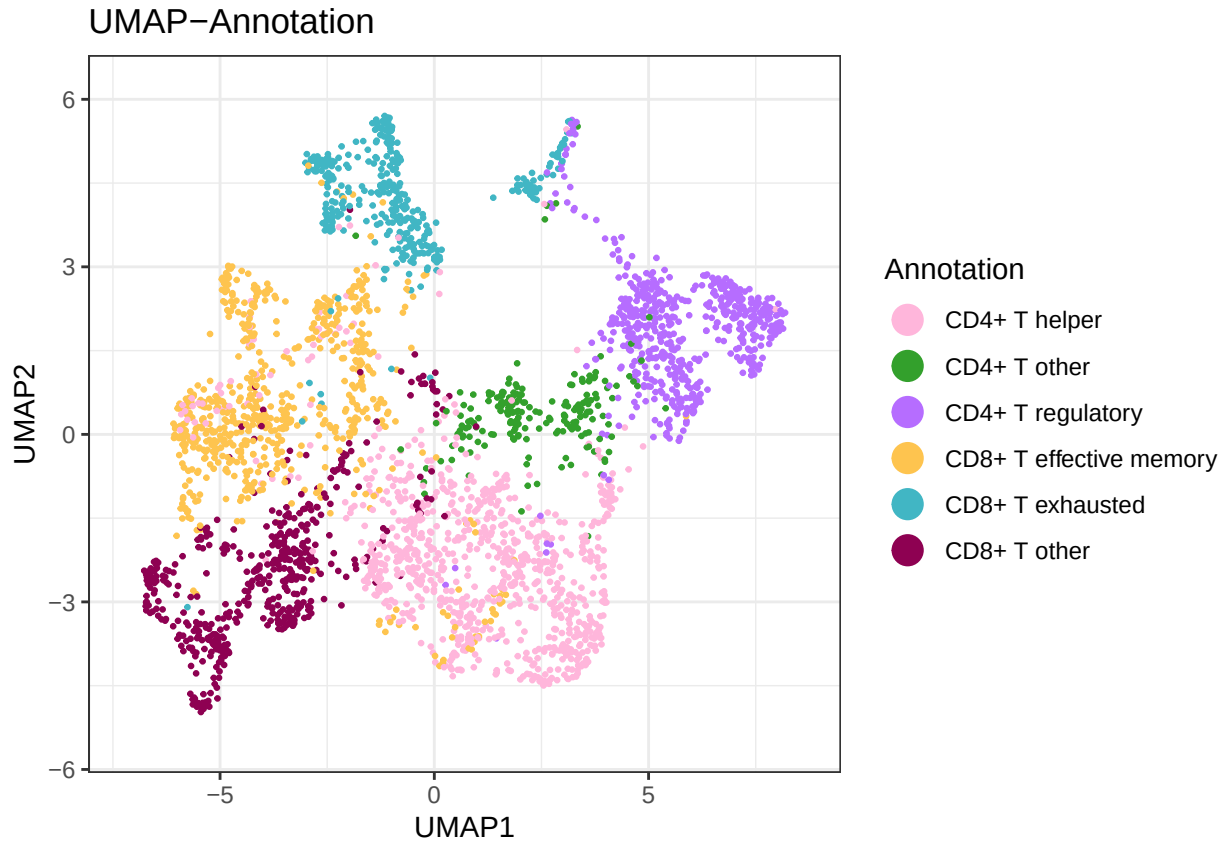


```
# plot
p2 <- ggplot(df, aes(x = UMAP1, y = UMAP2, color = T_type)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("UMAP-CD4+/CD8+")
p2
```



```
p3 <- ggplot(df, aes(x = UMAP1, y = UMAP2, color = Annotation)) +  
  geom_point(size = pointsize) +  
  guides(colour=guide_legend(override.aes=list(size = 5))) +  
  scale_color_manual(values=pal0) +  
  ylim(yrange) + xlim(xrange) +  
  ggtitle("UMAP-Annotation")
```

p3



```

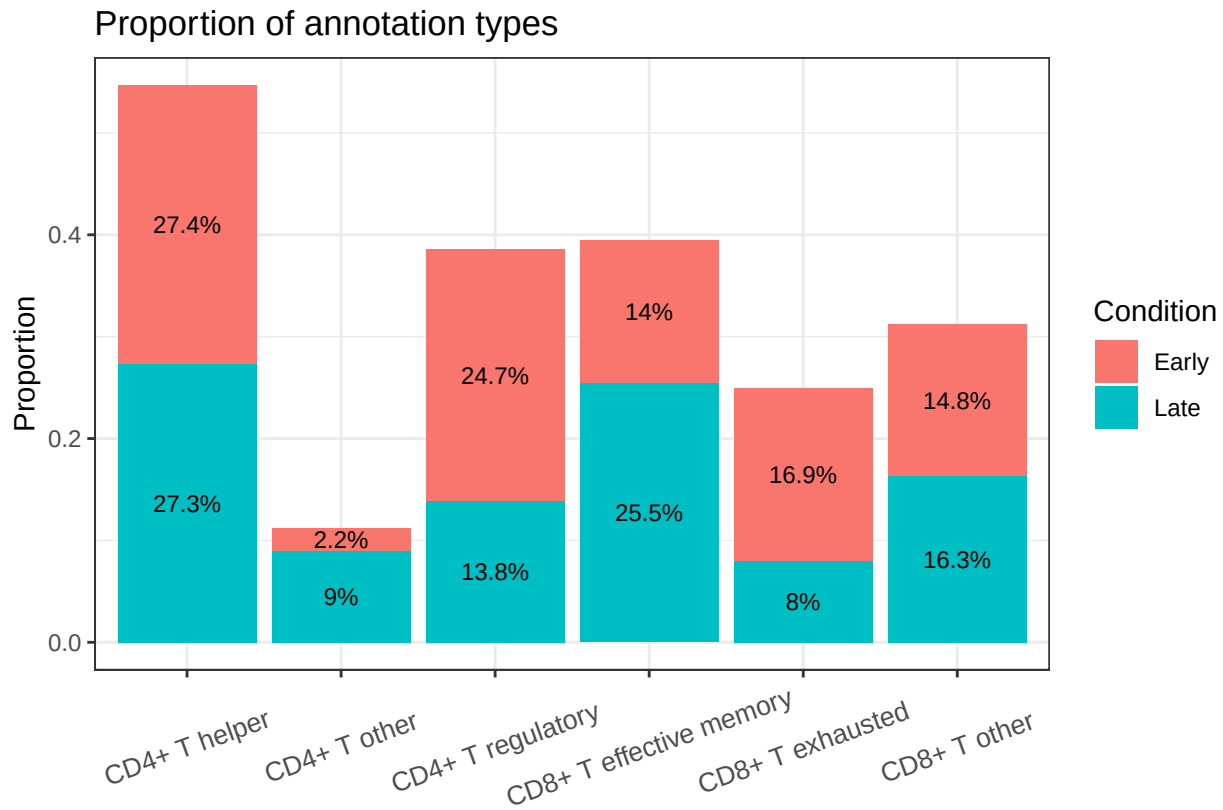
annot_L0 <- table(annotation_plot[which(onset == "LQCR")])
annot_Y0 <- table(annotation_plot[which(onset == "YQCR")])

annot_prop <- rbind(annot_L0/sum(annot_L0), annot_Y0/sum(annot_Y0))

n_annot <- ncol(annot_prop)
# plot
df_bar <- data.frame(Condition = c(rep("Late" , n_annot), rep("Early" , n_annot) ),
                      type = rep(colnames(annot_prop) , 2),
                      prop = c(annot_prop[1, ], annot_prop[2, ]))

p4 <- ggplot(df_bar, aes(fill=Condition, y=prop, x=type, label = paste0(round(prop, 3)*100, "%"))) +
  geom_bar(stat="identity") +
  scale_x_discrete(limits = annot_names) +
  geom_text(size = 3, position = position_stack(vjust = 0.5)) +
  ylab("Proportion") + xlab("") +
  theme(axis.text.x = element_text(angle = 20, vjust = 0.5, hjust=0.5, size = 10)) +
  ggtitle("Proportion of annotation types")
p4

```



```
plot <- ggarrange(p1, p2, ncol = 2, nrow = 1, align = "v", labels = c("(a)", "(b)"))
ggsave(filename = paste0(output_figure_dir, "/data.pdf"), width = 8, height = 3.5)

plot <- ggarrange(p3, p4, ncol = 2, nrow = 1, align = "v", labels = c("(c)", "(d)"))
ggsave(filename = paste0(output_figure_dir, "/data2.pdf"), width = 10, height = 3.5)
```

Run SASC model:

```
data_CD4 <- subset(x = input, subset = (T_cell_type == "CD4"))
out_CD4 <- run888model(data_CD4, H = 2, resolution = 0.3, fdr_thhd = 1e-60)
```

```
## Start pre-processing ...
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 1626
## Number of edges: 64097
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8504
## Number of communities: 6
## Elapsed time: 0 seconds
## Control FDR 1e-60 adding 33 biomarkers
## Finish pre-processing!
## MCMC iterations:
## 50 100 150 200 250 300 350 400 450 500 550 600 650
## Calculating the loss of the partitions...
##
## Got the point estimate of the partition!
```

```

##
## Extra mcmc for estimation of the weights.
## 50    100    150    200    250    300    350    400    450    500
data_CD8 <- subset(x = input, subset = (T_cell_type == "CD8"))
out_CD8 <- run888model(data_CD8, H = 2, resolution = 0.3, fdr_thhd = 1e-60)

## Start pre-processing ...
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 1513
## Number of edges: 63221
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8560
## Number of communities: 5
## Elapsed time: 0 seconds
## Control FDR 1e-60 adding 23 biomarkers
## Finish pre-processing!
## MCMC iterations:
## 50    100    150    200    250    300    350    400    450    500    550    600    650
## Calculating the loss of the partitions...
##
## Got the point estimate of the partition!
##
## Extra mcmc for estimation of the weights.
## 50    100    150    200    250    300    350    400    450    500
####
cell_names <- input@assays$RNA@counts@Dimnames[[2]]
T_type <- input@meta.data$T_cell_type
CD4_cells <- cell_names[which(T_type == "CD4")]
CD8_cells <- cell_names[which(T_type == "CD8")]

cluster <- rep(NA, length(cell_names))
names(cluster) <- cell_names
cluster[CD4_cells] <- out_CD4$cluster
n_clusters_CD4 <- max(as.numeric(out_CD4$cluster))
n_clusters_CD8 <- max(as.numeric(out_CD8$cluster))
n_clusters <- n_clusters_CD4 + n_clusters_CD8
cluster[CD8_cells] <- as.character(as.numeric(out_CD8$cluster) + n_clusters_CD4)

cluster_names <- as.character(1:n_clusters)
cluster_names

## [1] "1" "2" "3" "4" "5" "6" "7" "8" "9" "10" "11"

weights <- cbind(out_CD4$weights, out_CD8$weights)
cluster_type_CLE <- c(out_CD4$cluster_type_CLE, out_CD8$cluster_type_CLE)
colnames(weights) <- cluster_names

names(cluster_type_CLE) <- cluster_names

prob_in_h_mat_all <- matrix(NA, nrow = n_clusters, ncol = length(cell_names))
rownames(prob_in_h_mat_all) <- cluster_names
colnames(prob_in_h_mat_all) <- cell_names

```

```

prob_in_h_mat_all[1:n_clusters_CD4, CD4_cells] <- out_CD4$prob_in_h_mat_out
prob_in_h_mat_all[1:n_clusters_CD8, CD8_cells] <- out_CD8$prob_in_h_mat_out

xi_est <- cbind(out_CD4$xi_est[1:npc,], out_CD8$xi_est[1:npc,])
colnames(xi_est) <- cluster_names

weights_mat <- list()
weights_mat[[1]] <- cbind(out_CD4$weights_mat[[1]], out_CD8$weights_mat[[1]])
colnames(weights_mat[[1]]) <- cluster_names
weights_mat[[2]] <- cbind(out_CD4$weights_mat[[2]], out_CD8$weights_mat[[2]])
colnames(weights_mat[[2]]) <- cluster_names

```

Rename the clusters:

```

cluster_type_CLE_new <- c(cluster_type_CLE[order(match(cluster_type_CLE[1:n_clusters_CD4], c("C", "L",
cluster_type_CLE[(n_clusters_CD4 + 1):n_clusters][order(match(cluster_type_CLE[(n_clusters_CD4 + 1):n_clusters], c("C", "L",
cluster_names_new <- names(cluster_type_CLE_new)

cluster_new <- cluster
for (i in 1:n_clusters) {
  if (as.numeric(cluster_names_new[i]) != as.numeric(cluster_names[i])) {
    cluster_new[which(cluster == cluster_names_new[i])] <- as.numeric(cluster_names[i])
  }
}
weights_new <- weights[, cluster_names_new]
colnames(weights_new) <- cluster_names
weights_mat_new <- list()
for (i in 1:2) {
  weights_mat_new[[i]] <- weights_mat[[i]][, cluster_names_new]
  colnames(weights_mat_new[[i]]) <- cluster_names
}
xi_est_new <- xi_est[, cluster_names_new]
colnames(xi_est_new) <- cluster_names
prob_in_h_mat_all_new <- prob_in_h_mat_all[cluster_names_new, ]
rownames(prob_in_h_mat_all_new) <- cluster_names
names(cluster_type_CLE_new) <- cluster_names

cluster <- cluster_new
weights <- weights_new
xi_est <- xi_est_new
prob_in_h_mat <- prob_in_h_mat_all_new
cluster_type_CLE <- cluster_type_CLE_new
weights_mat <- weights_mat_new

```

Save data:

```

output <- input
output@meta.data$SASC_cluster <- cluster

clusterinfo <- list(cluster_type_CLE = cluster_type_CLE,
  xi_est = xi_est,
  prob_in_h_mat = prob_in_h_mat,
  weights = weights,

```

```

        weights_mat = weights_mat)
saveRDS(output, file = paste0(output_data_dir, "/SASC_output.rds"))
saveRDS(clusterinfo, file = paste0(output_data_dir, "/clusterinfo.rds"))

```

Visualization:

```

library(ggplot2); theme_set(theme_bw())
library(ggpubr)

# colors
pal <- c("#ffb6db", "#33A02C", "#b66dff", "#FEC44F", "#41B6C4", "#8E0152", "#0868AC", "#807DBA", "#E7298A",
        "#00441B", "#525252", "#4D9221", "#8B5742", "#D8DAEB", "#7cdd2d", "#980043", "#8C96C6", "#EC70C2",
        "#FDAE61", "#1D91C0", "#A6DBA0", "#4292C6", "#BF812D", "#01665E", "#41AB5D", "#FE9929", "#252525")
names(pal) <- 1:30

pointsize <- 0.5

weights <- clusterinfo$weights
weights_mat <- clusterinfo$weights_mat
cluster_type_CLE <- clusterinfo$cluster_type_CLE
prob_in_h_mat <- clusterinfo$prob_in_h_mat
cluster <- output@meta.data$SASC_cluster
annotation <- output@meta.data$annotation_final
gene <- output@reductions$pca@cell.embeddings
gene_umap <- output@reductions$umap@cell.embeddings
onset <- output@meta.data$Onset
cell_names <- output@assays$RNA@counts@Dimnames[[2]]
# rename the groups to be L and E
onset_LE <- onset
onset_LE[which(onset == "LOCRC")] <- "L"
onset_LE[which(onset == "YOCRC")] <- "E"
n_clusters <- length(unique(cluster))
cluster_names <- as.character(1:n_clusters)

#####
alpha <- 0.05
ci_low = c(apply(weights_mat[[1]], 2, quantile, alpha/2),
           apply(weights_mat[[2]], 2, quantile, alpha/2))

ci_high = c(apply(weights_mat[[1]], 2, quantile, (1 - alpha/2)),
            apply(weights_mat[[2]], 2, quantile, (1 - alpha/2)))

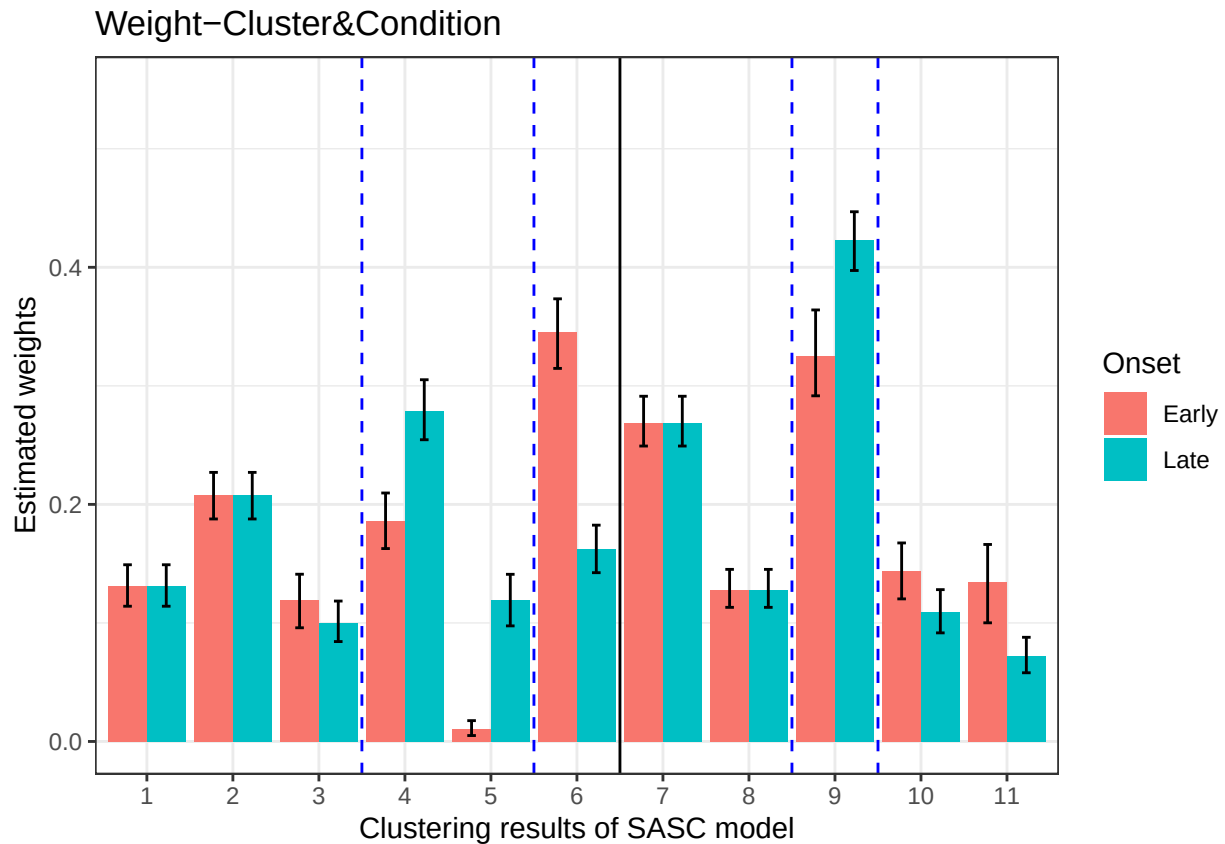
## Uncertainty of the weights
# plot
df_bar <- data.frame(Condition = c(rep("L" , n_clusters), rep("E" , n_clusters) ),
                    Onset = c(rep("Late" , n_clusters), rep("Early" , n_clusters) ),
                    Cluster = rep(as.character(1:(n_clusters)) , 2),
                    Weight = c(weights[1, ], weights[2, ]),
                    ci_low = ci_low,
                    ci_high = ci_high)

```

```

# mean and error bar
p1 <- ggplot(df_bar, aes(fill=Onset, y=Weight, x=Cluster)) +
  geom_bar(position="dodge", stat="identity") +
  scale_x_discrete(limits = cluster_names) +
  geom_errorbar(aes(ymin=ci_low, ymax=ci_high),
    width=.2,
    position=position_dodge(.9)) +
  geom_vline(xintercept = 6.5,
    color = "black", linewidth=0.5) +
  geom_vline(xintercept = 3.5, linetype="dashed",
    color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 5.5, linetype="dashed",
    color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 8.5, linetype="dashed",
    color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 9.5, linetype="dashed",
    color = "blue", linewidth=0.5) +
  ylim(c(0,0.55)) +
  xlab("Clustering results of SASC model") +
  ylab("Estimated weights") +
  ggtitle("Weight-Cluster&Condition")
p1

```



```

ggsave(paste0(output_figure_dir, "/weights.pdf"), height = 3.5, width = 4.3)

```

```
####
```

```

## num of different celltypes in each cluster

celltypes <- annotation
types <- names(table(celltypes))
pal0 <- pal[1:length(types)]
names(pal0) <- types

celltypes_annot <- types
num_celltypes <- length(celltypes_annot)
num_celltypes_cluster <- c()
for (ii in 1:(n_clusters)) {
  for (jj in 1:num_celltypes) {
    num_celltypes_cluster <- c(num_celltypes_cluster, length(which(celltypes[cluster == ii] == celltypes[jj])))
  }
}
cluster_temp <- c()
for (ii in 1:(n_clusters)) {
  cluster_temp <- c(cluster_temp, rep(ii, num_celltypes))
}

# plot
df_bar <- data.frame(Condition = rep(celltypes_annot , n_clusters),
                     Cluster = cluster_temp,
                     num_celltypes_cluster = num_celltypes_cluster)

####
## num of different celltypes in each clustertype

celltypes <- annotation
types <- names(table(celltypes))

celltypes_annot <- types
num_celltypes <- length(celltypes_annot)
num_celltypes_clustertype <- c()
clustertype <- rep(NA, length(cluster))
clustertype[which(cluster %in% which(cluster_type_CLE == "C"))] <- "C"
clustertype[which(cluster %in% which(cluster_type_CLE == "L"))] <- "L"
clustertype[which(cluster %in% which(cluster_type_CLE == "E"))] <- "E"
for (ii in unique(clustertype)) {
  for (jj in 1:num_celltypes) {
    num_celltypes_clustertype <- c(num_celltypes_clustertype, length(which(celltypes[clustertype == ii] == celltypes[jj])))
  }
}
cluster_temp <- c()
for (ii in unique(clustertype)) {
  cluster_temp <- c(cluster_temp, rep(ii, num_celltypes))
}

clustertype_names <- c("C_cd4T_hp", "C_cd4T_other", "C_cd4T_rg",
                     "L_cd4T_hp", "L_cd4T_other", "L_cd4T_rg",
                     "E_cd4T_hp", "E_cd4T_other", "E_cd4T_rg",
                     "C_CD8T_em", "C_CD8T_ex", "C_CD8T_other",

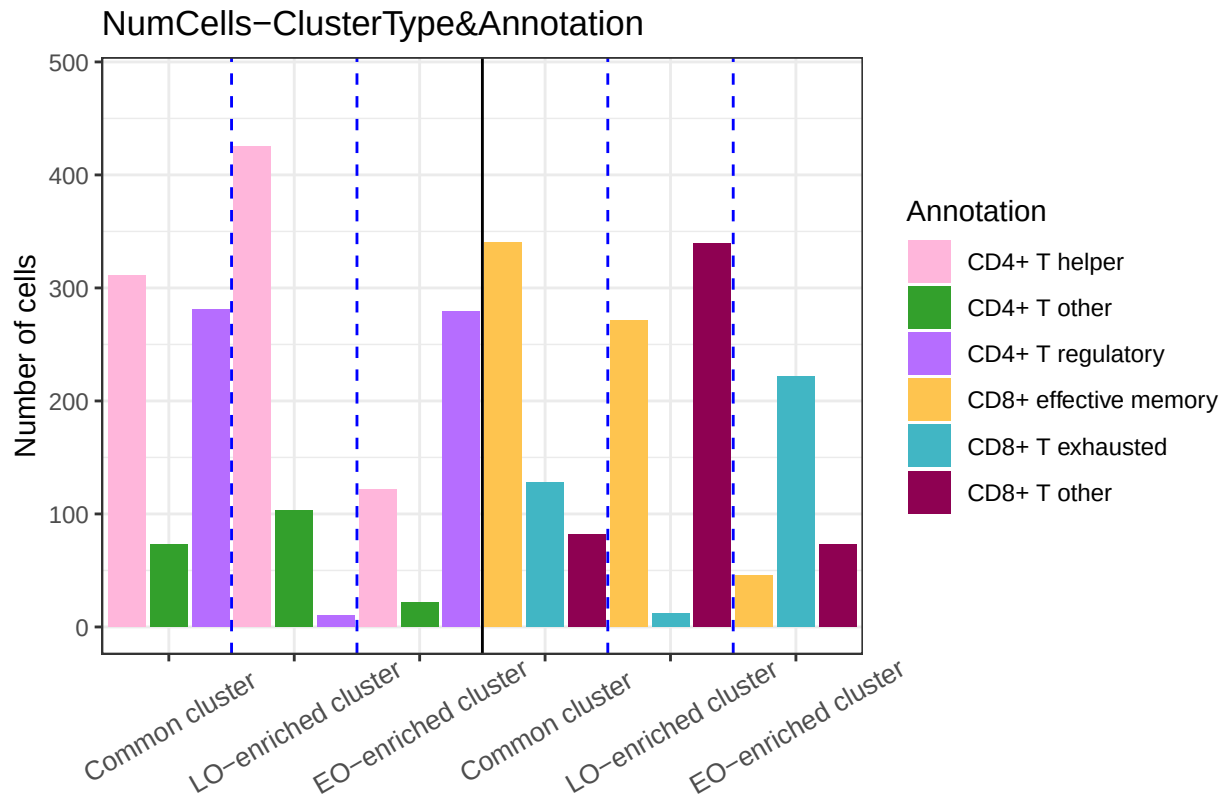
```

```

      "L_CD8T_em", "L_CD8T_ex", "L_CD8T_other",
      "E_CD8T_em", "E_CD8T_ex", "E_CD8T_other")

types_plot <- c("CD4+ T helper", "CD4+ T other", "CD4+ T regulatory",
               "CD8+ effective memory", "CD8+ T exhausted", "CD8+ T other")
pal0 <- pal[1:length(types_plot)]
names(pal0) <- types_plot
# plot
df_bar <- data.frame(Annotation = rep(types_plot, 3),
                    Cluster = paste0(cluster_temp, rep("_", length(cluster_temp)), rep(celltypes_annot
                    num_celltypes_clustertype = num_celltypes_clustertype)
df_bar$Cluster <- factor(df_bar$Cluster, levels = clustertype_names)
# Grouped bar plot
p3 <- ggplot(df_bar, aes(fill=Annotation, y=num_celltypes_clustertype, x=Cluster)) +
  geom_bar(stat="identity") +
  scale_fill_manual(values=pal0) +
  ylab("Number of cells") +
  theme(axis.text.x = element_text(angle = 30, vjust = 0.6, hjust=0.5, size = 10)) +
  geom_vline(xintercept = 9.5,
            color = "black", linewidth=0.5) +
  geom_vline(xintercept = 3.5, linetype="dashed",
            color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 6.5, linetype="dashed",
            color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 12.5, linetype="dashed",
            color = "blue", linewidth=0.5) +
  geom_vline(xintercept = 15.5, linetype="dashed",
            color = "blue", linewidth=0.5) +
  ylim(c(0,480)) + xlab("") +
  scale_x_discrete(breaks=c("C_cd4T_other", "L_cd4T_other", "E_cd4T_other",
                           "C_CD8T_ex", "L_CD8T_ex", "E_CD8T_ex"),
                  labels=c("Common cluster", "LO-enriched cluster", "EO-enriched cluster",
                           "Common cluster", "LO-enriched cluster", "EO-enriched cluster")) +
  ggtitle("NumCells-ClusterType&Annotation")
p3

```

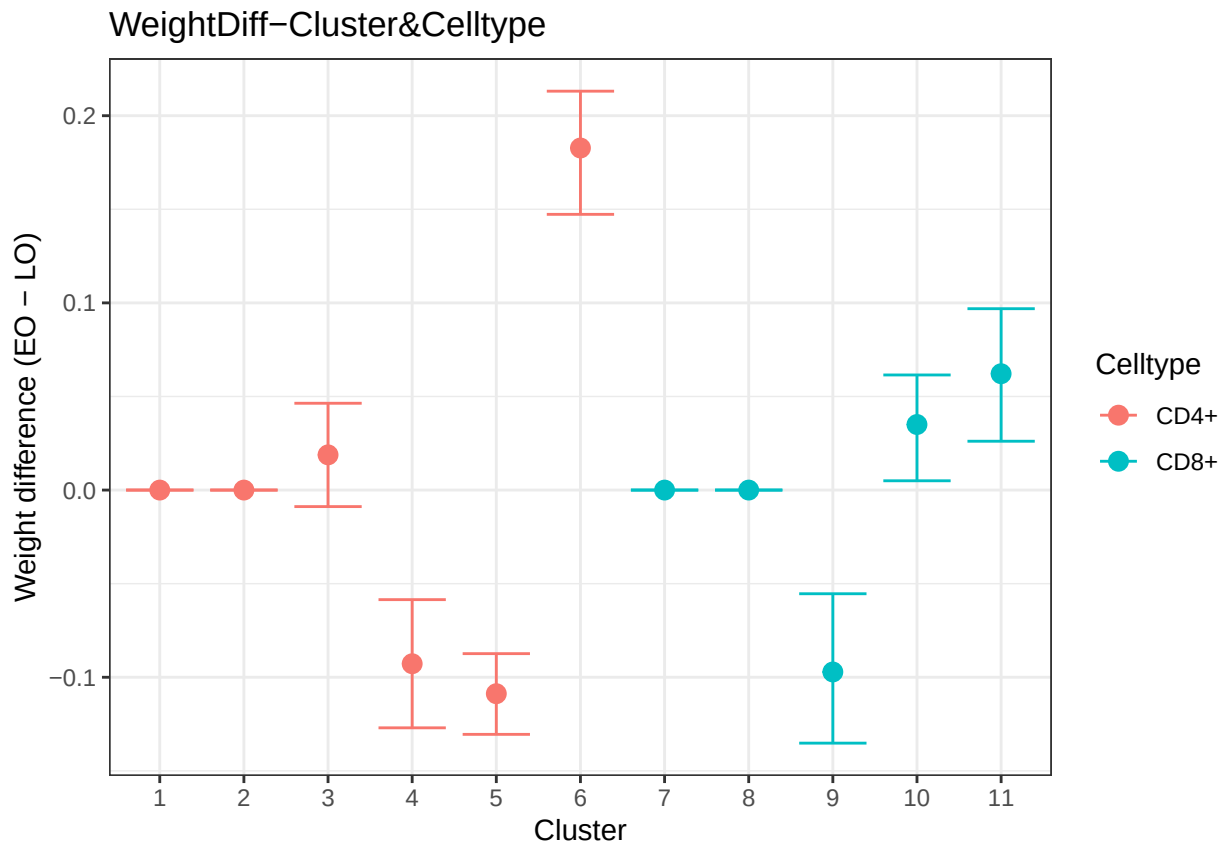


```
ggsave(paste0(output_figure_dir, "/num_celltypes_clustertype.pdf"), height = 4, width = 5.5)

##### Annotate cluster types
num_celltypes_cluster_mat <- matrix(num_celltypes_cluster, ncol = num_celltypes, byrow = TRUE)
node_annot <- rep(NA, n_clusters)
cell_annot <- rep(NA, length(cell_names))
for (i in 1:n_clusters) {
  node_annot[i] <- types[which.max(num_celltypes_cluster_mat[i, ])]
  cell_annot[which(cluster == as.character(i))] <- node_annot[i]
}

cluster_annot <- rep(NA, n_clusters)
for (i in 1:n_clusters) {
  cluster_annot[i] <- node_annot[i]
}
names(cluster_annot) <- cluster_names <- as.character(1:(n_clusters))
# node_annot is the cluster annotations
# cell_annot is the cell annotations (the ones corresponding to the cluster assigned to)
df <- data.frame(Cluster = as.character(1:(n_clusters)),
  weightLO_EO = -apply(weights_mat[[1]] - weights_mat[[2]], 2, mean),
  ci_low = -apply(weights_mat[[1]] - weights_mat[[2]], 2, quantile, alpha/2),
  ci_high = -apply(weights_mat[[1]] - weights_mat[[2]], 2, quantile, (1 - alpha/2)),
  Celltype = c(rep("CD4+", 6), rep("CD8+", 5)))
p2 <- ggplot(df, aes(x=Cluster, y= weightLO_EO)) +
  geom_errorbar(width=0.8, aes(ymin=ci_low, ymax=ci_high, col = Celltype)) +
  # scale_color_manual(values=pal0) +
  geom_point(aes(col = Celltype), size = 3) +
```

```
scale_x_discrete(limits = cluster_names) +
ylab("Weight difference (EO - LO)") +
ggtitle("WeightDiff-Cluster&Celltype")
p2
```



```
ggsave(paste0(output_figure_dir, "/weightsLO_EO_color.pdf"), height = 4, width = 5)
```

```
#####
### plot uncertainty
H <- 2

h_temp <- H + 3 # the cluster of interest
prob_in_h <- prob_in_h_mat[h_temp,]

df <- data.frame(PCA1 = gene[,1], PCA2 = gene[,2],
                 UMAP1 = gene_umap[,1], UMAP2 = gene_umap[,2],
                 Onset = onset_LE, cluster = cluster, Similarity = prob_in_h)

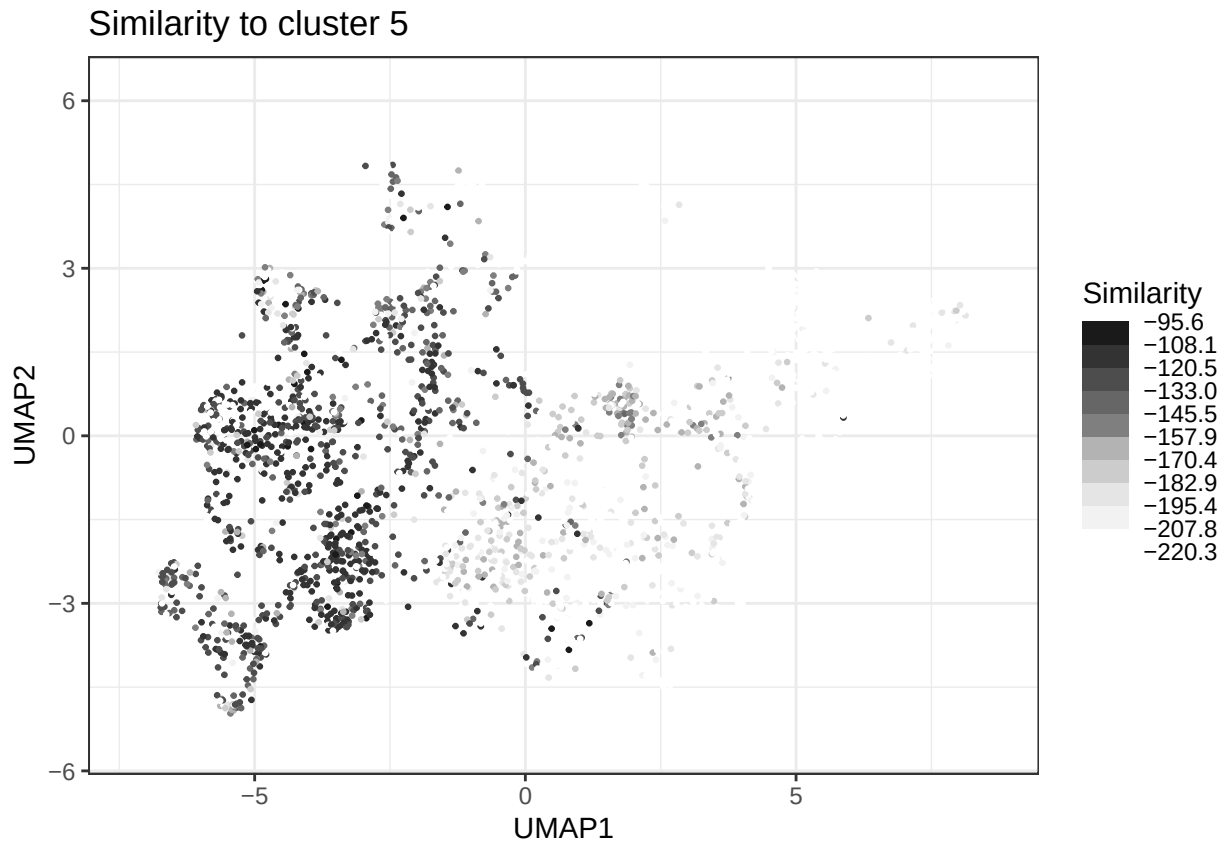
xrange <- c(min(df$UMAP1) - 0.5, max(df$UMAP1) + 0.5)
yrange <- c(min(df$UMAP2) - 0.5, max(df$UMAP2) + 0.5)

# df_naomit <- na.omit(df)
lim_lower <- round(as.numeric(quantile(na.omit(prob_in_h), 0.5)), 1) - 0.1
lim_upper <- round(max(na.omit(prob_in_h)), 1) + 0.1
p_uncert <- ggplot(data=df) +
  geom_point(aes(UMAP1, UMAP2, colour = Similarity), size=pointsizes) +
```

```

binned_scale(aesthetics = "color",
             scale_name = "stepsn",
             palette = function(x) c("grey100", "grey95", "grey90", "grey80", "grey70", "grey50", "grey30", "grey10", "white"),
             breaks = round(seq(lim_lower, lim_upper, length.out = 11), 1),
             limits = c(lim_lower, lim_upper),
             show.limits = TRUE,
             guide = "colorsteps") +
ylim(yrange) + xlim(xrange) +
ggtitle(paste0("Similarity to cluster ", h_temp))
p_uncert

```



```

ggsave(paste0(output_figure_dir, "/Prob.pdf"), height = 4, width = 5)

```

```

#####
# plot in UMAP score

pointsize <- 0.5

# colors
pal <- pal[1:(n_clusters)]

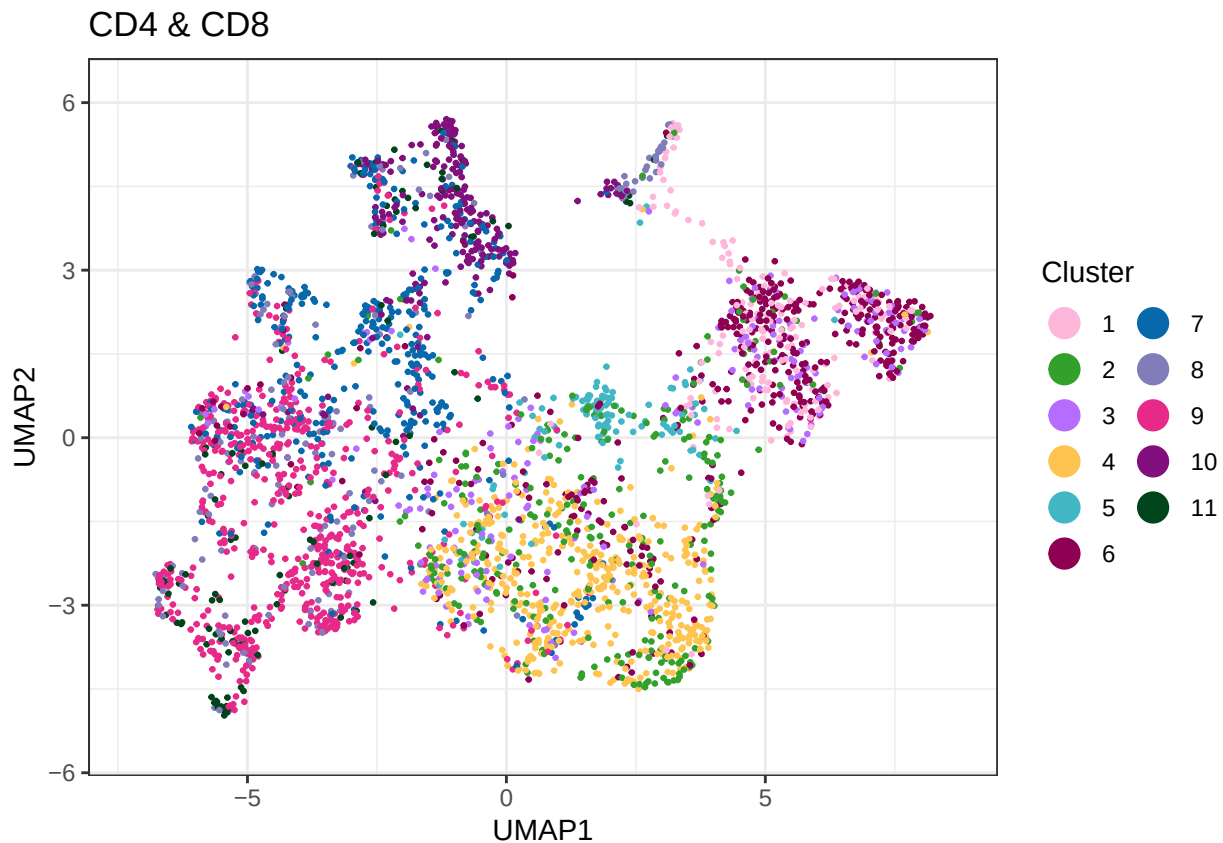
#####
df <- data.frame(PCA1 = gene[,1], PCA2 = gene[,2],
                 UMAP1 = gene_umap[,1], UMAP2 = gene_umap[,2],
                 Onset = onset_LE, Cluster = cluster)

```

```
df$Cluster <- factor(df$Cluster, levels = as.character(1:n_clusters))

df_naomit <- na.omit(df)

p1_cluster <- ggplot(df_naomit, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5), ncol = 2)) +
  scale_color_manual(values=c(pal)) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("CD4 & CD8")
p1_cluster
```

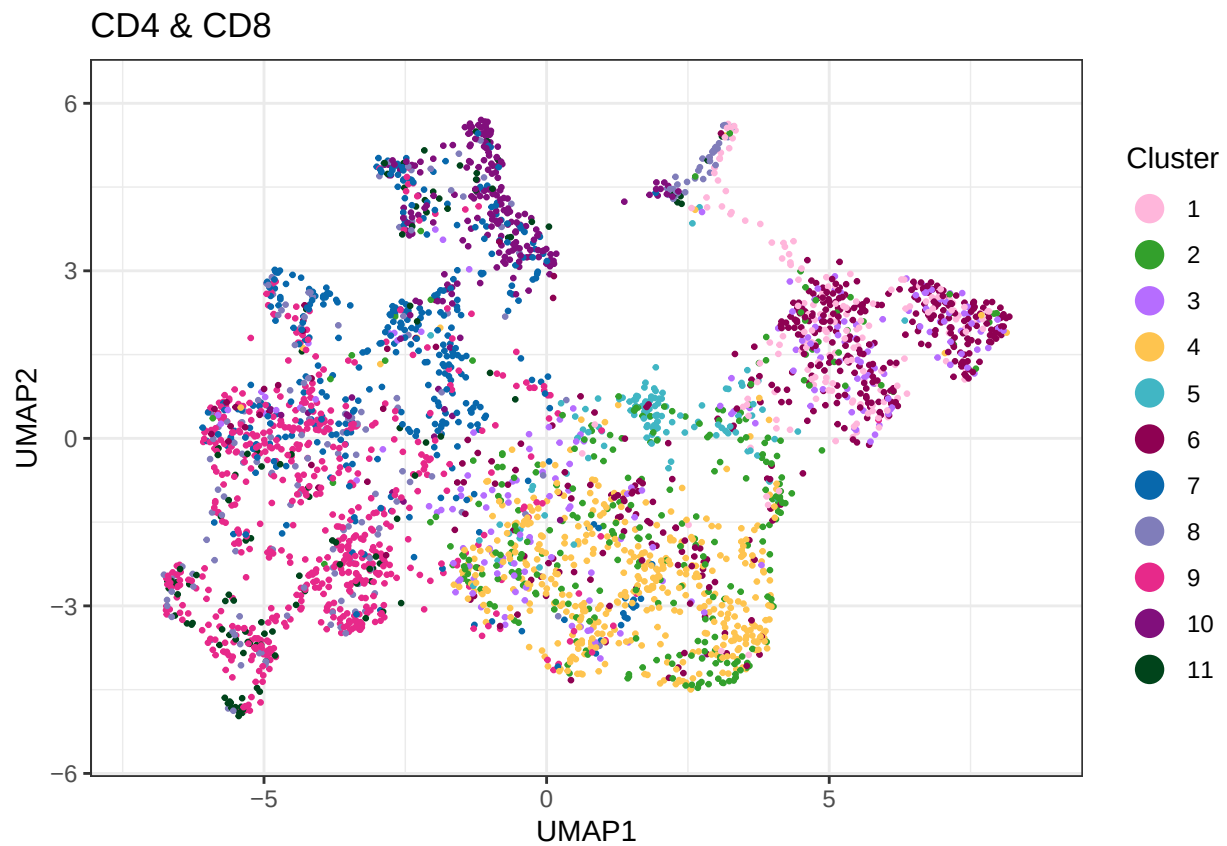


```
#####
# UMAP plots

df <- data.frame(PCA1 = gene[,1], PCA2 = gene[,2],
                 UMAP1 = gene_umap[,1], UMAP2 = gene_umap[,2],
                 Onset = onset_LE, Cluster = cluster)
df$Cluster <- factor(df$Cluster, levels = as.character(1:n_clusters))
df_naomit <- na.omit(df)

p1 <- ggplot(df_naomit, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 4.5))) +
  scale_color_manual(values=c(pal)) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("CD4 & CD8")
```

p1



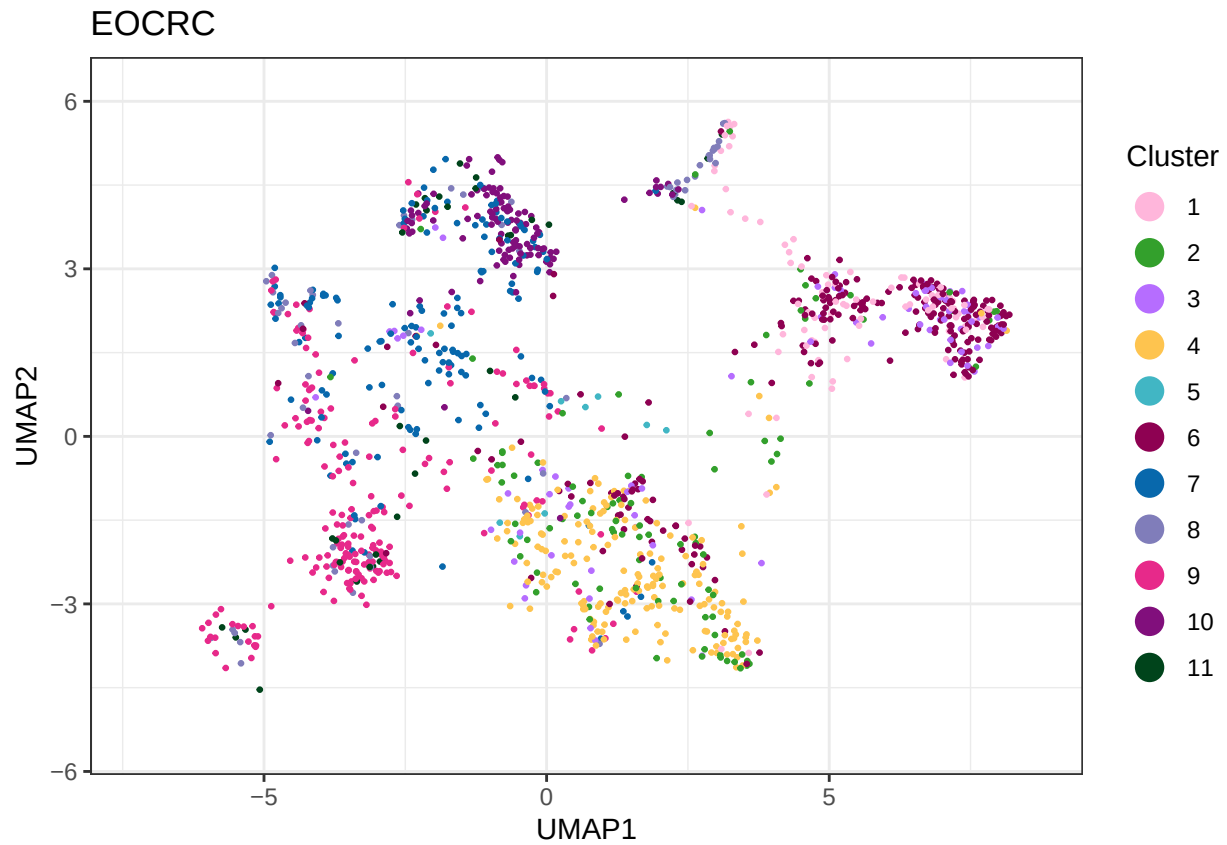
```
# plot cluster comparison
```

```
df_YOCRC <- df_naomit[which(df_naomit$Onset == "E"), ]
```

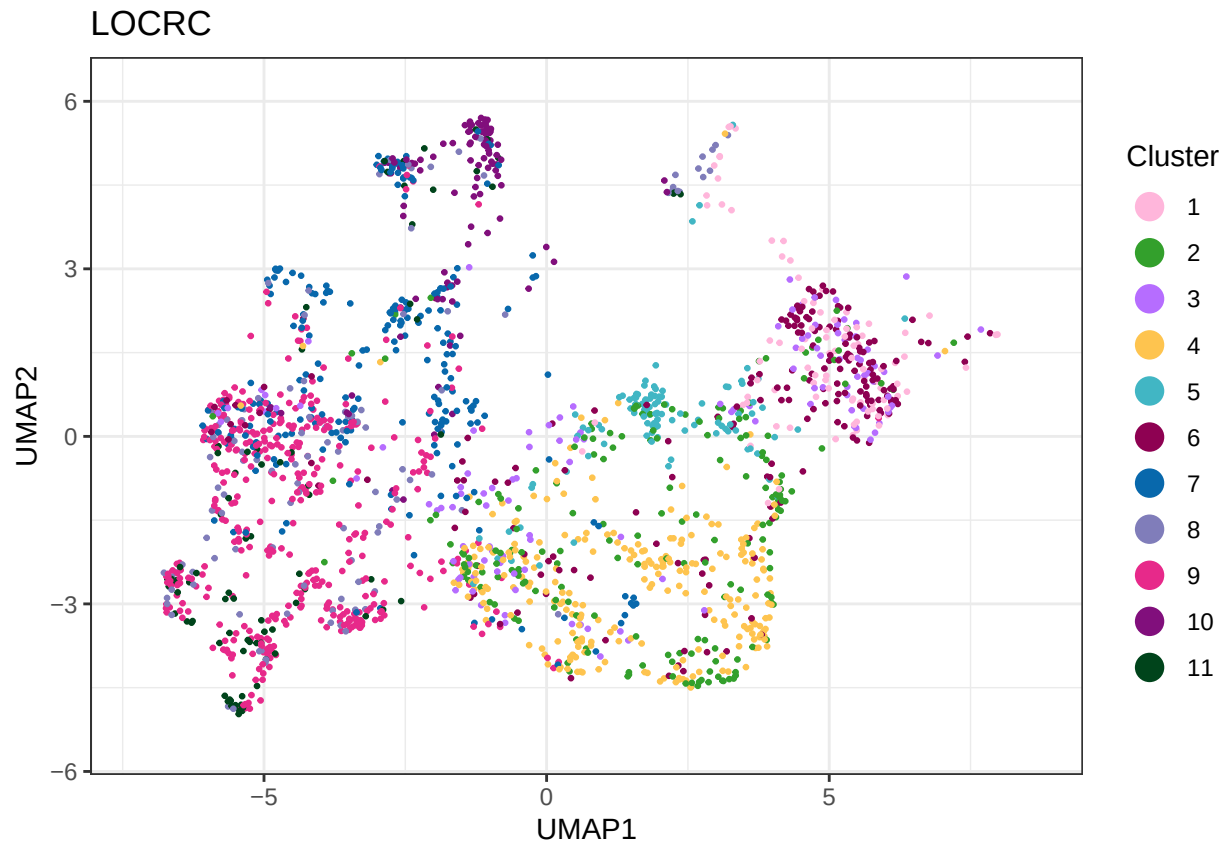
```
# plot
```

```
p2 <- ggplot(df_YOCRC, aes(x = UMAP1, y = UMAP2, color = Cluster)) +  
  geom_point(size = pointsize) +  
  guides(colour=guide_legend(override.aes=list(size = 4.5))) +  
  scale_color_manual(values=c(pal)) +  
  ylim(yrange) + xlim(xrange) +  
  ggtitle("EOCRC")
```

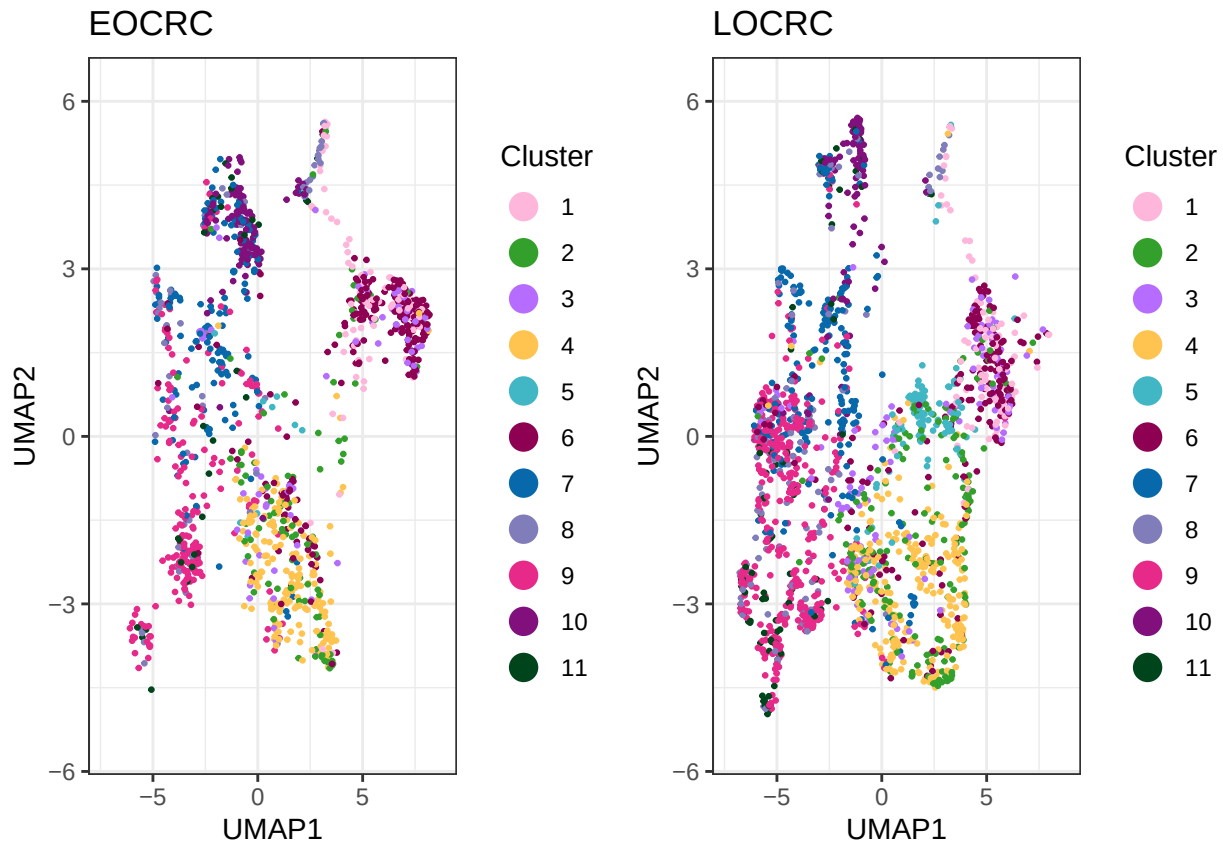
p2



```
df_LOCRC <- df_naomit[which(df_naomit$Onset == "L"), ]
# plot
p3 <- ggplot(df_LOCRC, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 4.5))) +
  scale_color_manual(values=c(pal)) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("LOCRC")
p3
```



```
plot <- ggarrange(p2, p3, ncol = 2, nrow = 1, align = "hv")  
plot
```



```
ggsave(filename = paste0(output_figure_dir, "/umap_all12.pdf"), width = 7.5, height = 3.5)

#####

Y0cluster_ind <- which(cluster_type_CLE == "E")
## Y0cluster
df_Y0CRC_Y0cluster <- df_naomit[which(df_naomit$Onset == "E" &
                                     as.numeric(df_naomit$Cluster) %in% Y0cluster_ind ), ]

# plot
p1 <- ggplot(df_Y0CRC_Y0cluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[Y0cluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("EOCRC EOcluster")

df_LOCRC_Y0cluster <- df_naomit[which(df_naomit$Onset == "L" &
                                     as.numeric(df_naomit$Cluster) %in% Y0cluster_ind ), ]

# plot
p2 <- ggplot(df_LOCRC_Y0cluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[Y0cluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("LOCRC EOcluster")

L0cluster_ind <- which(cluster_type_CLE == "L")
```

```

## LOcluster
df_YOCRC_LOcluster <- df_naomit[which(df_naomit$Onset == "E" &
                                     as.numeric(df_naomit$Cluster) %in% LOcluster_ind), ]

# plot
p3 <- ggplot(df_YOCRC_LOcluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[LOcluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("EOCRC LOcluster")

df_LOCRC_LOcluster <- df_naomit[which(df_naomit$Onset == "L" &
                                     as.numeric(df_naomit$Cluster) %in% LOcluster_ind), ]

# plot
p4 <- ggplot(df_LOCRC_LOcluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[LOcluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("LOCRC LOcluster")

## COMMONcluster
COcluster_ind <- which(cluster_type_CLE == "C")

df_YOCRC_COMMONcluster <- df_naomit[which(df_naomit$Onset == "E" &
                                     as.numeric(df_naomit$Cluster) %in% COcluster_ind), ]

# plot
p5 <- ggplot(df_YOCRC_COMMONcluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[COcluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("EOCRC COMMONcluster")

df_LOCRC_COMMONcluster <- df_naomit[which(df_naomit$Onset == "L" &
                                     as.numeric(df_naomit$Cluster) %in% COcluster_ind), ]

# plot
p6 <- ggplot(df_LOCRC_COMMONcluster, aes(x = UMAP1, y = UMAP2, color = Cluster)) +
  geom_point(size = pointsize) +
  guides(colour=guide_legend(override.aes=list(size = 5))) +
  scale_color_manual(values=c(pal[COcluster_ind])) +
  ylim(yrange) + xlim(xrange) +
  ggtitle("LOCRC COMMONcluster")

plot <- ggarrange(p5, p6, p3, p4, p1, p2, ncol = 2, nrow = 3, align = "v")
ggsave(filename = paste0(output_figure_dir, "/umap_clusters.pdf"), width = 7.5, height = 9)

```